Hackathon Week 3 - Resit

Project title: Healthcare Easy

My pair partner: Slyman Arnaot

Tool we had to use: Claude API or OpenAI API

SDG we had to address: SDG 10 - Reduced Inequalities

What problem does it solve, and for whom?

Healthcare Easy is made for people in the Netherlands who have trouble understanding Dutch healthcare messages. Our main target group is patients who do not speak Dutch very well and find it difficult to understand medical information. Our second target group is people who do speak Dutch but have trouble understanding difficult medical words.

Specific user situation

For example, a patient receives a message from a Dutch hospital about an appointment. The patient is at home and reads the message, but does not understand some medical words or instructions. They want to know what the message means, what they need to do and what information is important.

The patient needs a simple explanation, preferably in a language they understand better. It is important that the meaning of the original message stays the same.

Why is this a problem?

Not everyone can understand healthcare information as easily. People who do not speak Dutch very well or have trouble understanding medical information may find it harder to understand messages from doctors or hospitals.

This can cause problems because they might miss important information or misunderstand what they need to do.

Ethical reflection

The biggest risk of Healthcare Easy is that AI could explain important medical information incorrectly. For example, it could change or leave out information about medication, dates or warnings. This could be dangerous for people who do not speak Dutch very well, because they might not notice that the AI made a mistake.

According to a study by RIVM and Nivel from 2026, with 11,441 participants, 52.8% had trouble with finding, understanding, checking or using health information. This shows that understanding healthcare information is already difficult for many people.

Another risk is that AI might work better in some languages than in others. This could create new inequalities instead of reducing them.

To reduce these risks, our tool should only explain the information from the original message and not give medical advice. Important information, like dates, medication and warnings, should stay the same. If something is unclear, users should check with a doctor or another healthcare professional.

We also have not tested our tool with real patients yet, so we do not know for sure if all the explanations are correct and safe.

Source: RIVM & Nivel (2026): https://www.nivel.nl/nl/publicatie/gezondheidsvaardigheden-nederland-methoden-en-resultaten-van-de-m-pohl-health-literacy-0

Code explanations:

Python code:

```
113    The goal is:
114
115    TRANSLATE + EXPLAIN + SIMPLIFY
116
117    The original healthcare message is the source of truth.
```

Explanation:

We made the prompt based on the problem of our target group. Our research showed that translating healthcare messages is not always enough, because people can still have trouble understanding medical words. That is why we gave the AI three main tasks: translate, explain and simplify. We want the AI to explain the message in easy language, but without changing the original meaning.

Python code:

```
1. DO NOT diagnose diseases or medical conditions.
2. DO NOT prescribe medication.
3. DO NOT recommend treatment.
4. DO NOT give new medical instructions.
5. DO NOT invent medical information.
6. DO NOT guess information that is missing.
7. DO NOT change important medical information.
```

Explanation:

We also added safety instructions to our prompt because we are working with healthcare information. For example, the AI is not allowed to give medical advice or make up information. We added these rules because our target group might not notice if the AI makes a mistake. This could be dangerous, especially with medication or other important instructions. These rules should reduce the risk, but they do not guarantee that the AI will always give a correct explanation.

Python code:

```
8. If the original message is unclear, say that it is unclear.
9. If important information is missing, identify the missing information.
10. If clarification is needed, say that the user should ask an appropriate
healthcare professional for clarification.

Do NOT provide the missing information yourself.
```

Explanation:

Another important choice was telling the AI not to guess when information is missing. For example, if a hospital message does not explain when a patient should stop eating before a blood test, the AI should not make up a time. Instead, it should say that the information is missing and that the patient should ask a healthcare professional. We chose this because giving wrong information could be worse than saying something is unclear.

How does our tool connect to our user group?

When we made Healthcare Easy, we looked at the problems of our user group. Some patients do not understand difficult medical words. That is why we told the AI to explain messages in easy words instead of only translating them. We also added sections that show what the patient needs to do and what information is important, like dates and warnings. Because some patients do not speak Dutch very well, we

added five different languages. We also added safety rules to the AI, so it should not change important information or give medical advice. Our goal is to help people understand healthcare messages better, even if they do not speak Dutch very well.